

Error Calculation Report

● Assignment 01 – AI/ML for Mobility Applications

1. Introduction

This document presents the error analysis between the actual travel distance obtained from Google Maps and the calculated distance derived from the recorded velocity data. The velocity values were recorded at 30-second intervals during the journey.

2. Given Data

Time interval (Δt) = 30 seconds

Velocity unit = km/h

Actual Distance (from Google Maps) = **13 km**

3. Method Used for Calculation

The distance was calculated using the basic kinematic relation:

Distance = Velocity $\times$ Time

Since velocity is given in km/h, the time interval was converted from seconds to hours:

$$30 \text{ Sec} = [30/3600] \text{ hours}$$

For each velocity reading:

$$\text{Distance} = \text{Velocity} \times 0.00833$$

The total distance was obtained by summing all individual distances.

4. Results

Calculated Distance (From Velocity Data): 12.4 km

Actual Distance (From Google Maps): 13 km

5. Error Calculation

$$\text{Absolute Error: Error} = | \text{Actual} - \text{Calculated} | = | 13 - 12.4 | = 0.6 \text{ km}$$

$$\text{Percentage Error} = (| \text{Actual} - \text{Calculated} | / \text{Actual}) \times 100 = [0.6 / 13] \times 100 = 4.62\%$$

6. Conclusion

The calculated distance (12.4 km) is close to the actual distance (13 km). The percentage error obtained is 4.62%.